TECHNICAL OVERVIEW & FEATURE GUIDE

LogiCoy eRx for OpenEMR

Electronic prescribing from inside the patient chart: how the integration works, what it does, and what your practice needs to run it.

Prepared for client practices and IT teams
OpenEMR module oe-module-logicoy-erx · version 1.0.7

September 2026
erx.logicoy.com

LogiCoy

California-based corporation
Founded in 2009
Specialists in PDMP, integrations and electronic prescribing

LogiCoy eRx

Prescribe in seconds. Decide with confidence. With a team that actually picks up the phone, at a price that scales with you.

STREAMLINED, SECURE AND COMPLIANT

LogiCoy eRx is a high-performance electronic prescribing platform that simplifies clinical workflows while keeping you fully compliant with regulations. From independent practitioners to large EHR vendors, LogiCoy provides a secure, efficient path from clinical decision to pharmacy fulfillment.

WHY LOGICOY ERX

EPCS Certified
Prescribe controlled substances with integrated identity proofing.

Clinical Insights
Patient medication history and real-time eligibility.

Efficiency Tools
Electronic Prior Authorization (ePA) and Real-Time Pharmacy Benefits (RTPB).

Secure & Compliant
Fully HIPAA compliant with robust data protection.

★★★★★
LogiCoy eRx is a low-cost, customizable solution that was ideal for my practise. Importing patient records was quick and easy.

California Dentist

WHY LOGICOY

Complete Functionality
A true end-to-end patient care workflow.

Fully Certified
Stay ahead of regulatory changes on a platform that stays compliant.

Cost-Effective
Enterprise-grade features without the enterprise-grade price tag.

Reliable Support
Toll-free phone support whenever you need help.

★★★★★
You made it all so simple, reliable, and easy. The LogiCoy team helped us build our technology engine that is helping countless patients across the United States. Thank you.

Pejman Salimpour, M.D., F.A.A.P. · CEO, Bluestone Health

★★★★★
I truly enjoy the software and service you guys have provided.

Dr. Hollis Clark, M.D. · Internal Medicine

FOUR WAYS TO CONNECT: CHOOSE THE INTEGRATION LEVEL THAT FITS

01
PLATFORM ACCESS
Direct portal access. Sign up and start sending prescriptions in under 15 minutes.

Best for: independent prescribers

THIS GUIDE

02
EMBEDDED WEB APP
An eRx button inside your EHR/EMR with single sign-on and automated messaging.

Best for: rapid EHR integration

03
ERX APIS
Core and Advanced APIs to build a native prescribing experience in your own UI.

Best for: custom EHR development

04
FULL INTEGRATION
Bespoke coding and functionality tailored to your platform's requirements.

Best for: fully bespoke solutions

Book a demo
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LogiCoy

ABOUT THIS DOCUMENT

LogiCoy eRx inside OpenEMR

This guide is for practice owners, clinical leads and IT staff evaluating or deploying LogiCoy eRx on OpenEMR. It covers what the integration does for clinicians, how it is built, the security model, and what your server needs.

The installation walkthrough is in the separate **LogiCoy eRx for OpenEMR: Client Setup Guide**, which LogiCoy supplies with your onboarding pack.

At a glance

1 click
from the Patient Summary to the prescribing screen

0
inbound firewall ports; the module only makes outbound HTTPS calls

0
OpenEMR core files changed; installs as a standard custom module

7.0+
OpenEMR version supported, verified against 8.2.0

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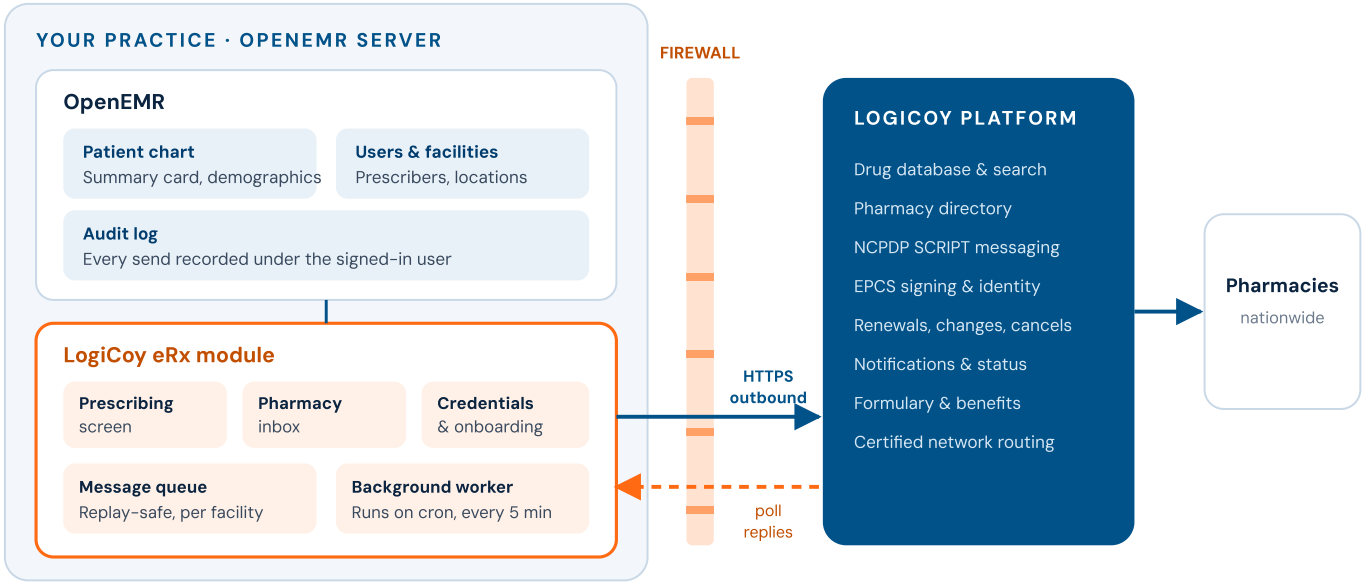
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02 · ARCHITECTURE

How the integration works

LogiCoy eRx for OpenEMR is an **embedded integration** (option 02 of LogiCoy's four ways to connect). A small module installed in your OpenEMR connects the patient chart to the LogiCoy network. The clinical work (the drug database, prescription messaging, controlled-substance signing and pharmacy routing) runs on LogiCoy's certified platform.



IN YOUR OPENEMR

The LogiCoy eRx module

- Prescribing card and screen in patient context
- Pharmacy request inbox and prescription log
- Encrypted practice and prescriber credentials
- Outbound queue and background worker
- Tracks every sent prescription and its status

AT LOGICOY

The certified eRx platform

- Drug database, controlled-substance flags, formulary
- NCPDP SCRIPT message construction
- EPCS signing and PKI
- Routing to pharmacies across the network
- Regulatory updates, applied centrally

Why it is split this way. The module has no clinical logic of its own, so it stays small, easy to audit and quick to update when OpenEMR releases a new version. Certification-sensitive work stays on LogiCoy's platform, where changes reach every practice at once without a module upgrade.

03 · FEATURES FOR CLINICIANS

Prescribing in the patient chart

One-click launch from Patient Summary

A **LogiCoy eRx** card on the Patient Summary opens the prescribing screen with the patient already loaded. Only users with OpenEMR's prescribing permission see it.

Drug search by name or NDC

Type a product name or paste an NDC to search LogiCoy's drug database. Controlled substances are flagged in the results.

Complete prescription entry

Directions (sig), quantity, days supply, refills, start date and notes to the pharmacist, with required fields checked before saving.

Pharmacy directory search

Find the patient's pharmacy by name, city or ZIP from the national directory, then attach it to the prescription.

Drafts, then send

Save, edit or remove drafts before sending. **Send** transmits the new prescription (NewRx) through LogiCoy to the chosen pharmacy.

Sent prescriptions with live status

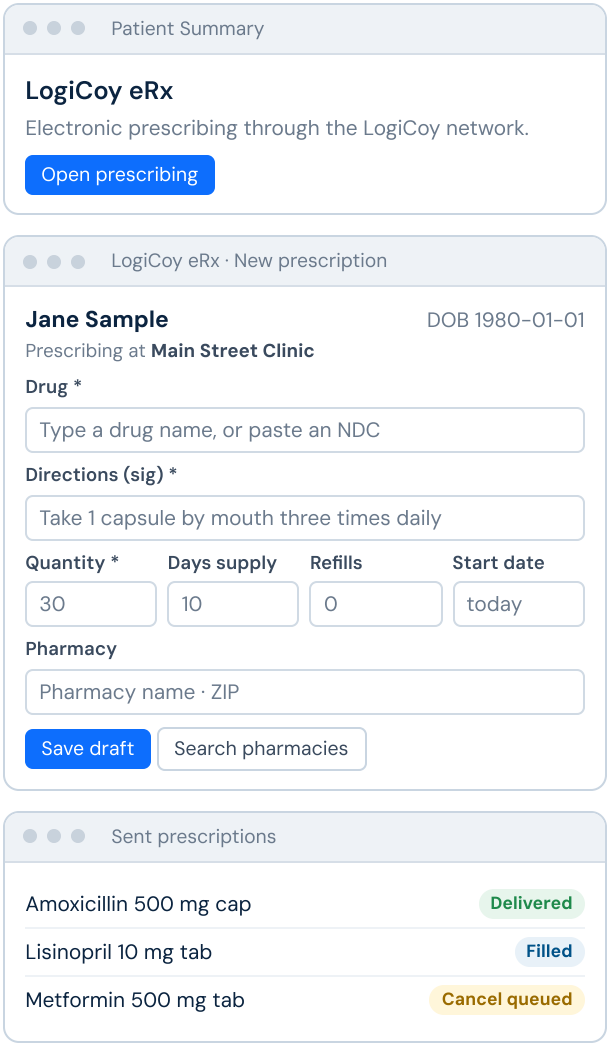
Every sent prescription is listed with its current status. **Check status** fetches the latest delivery, fill and verification state on demand.

Tracked and audited

Every sent prescription is kept with the patient for status, cancel and pharmacy follow-up, and recorded in OpenEMR's audit log under the signed-in prescriber.

Controlled substances (EPCS)

Prescribers registered as EPCS can prescribe controlled substances. Signing and identity proofing are handled by LogiCoy's EPCS-certified platform.



Illustrative screens with synthetic data. Layout follows your OpenEMR theme.

04 · FEATURES FOR CLINICIANS AND STAFF

Pharmacy requests, cancellations and status

Pharmacy requests reach OpenEMR automatically. Prescribers answer them in the chart or the inbox.

RXRENEWAL

Refill and renewal requests

When a pharmacy asks to renew a prescription, the request shows on the patient card and in the inbox with its full details. The prescriber **approves** it, or **denies** it with a standard reason code. The response is queued and sent on the next worker run.

RXCHANGE

Change requests

Pharmacy requests to change a medication (for example a therapeutic substitution or prior-authorization change) can be **approved**, **validated** or **denied** for one requested medication. The coded medication supplied by the platform is sent back unchanged.

RXCANCEL

Cancel a sent prescription

Cancel from the **Sent prescriptions** list. Only prescriptions that can still be cancelled offer the option, and one cancel per prescription is allowed. The pharmacy's answer (approved, denied or not delivered) updates the status automatically.

STATUS

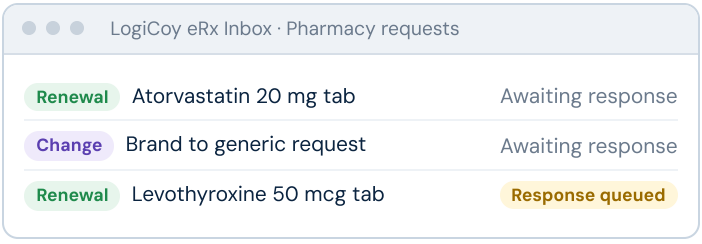
Delivery, fill and verification tracking

Prescription status, fill status and pending-verification status are fetched on request, and again automatically whenever LogiCoy reports a fill or verification event.

Practice-wide pharmacy inbox

Modules → LogiCoy eRx Inbox lists every renewal and change request waiting for a response at the facility, with its priority and state. Anyone with prescribing permission can see the list; a request can only be answered by the prescriber it is addressed to.

- A request that has already been answered cannot be answered twice
- Requests are matched to the patient's chart when the patient is known
- New requests arrive each time the background worker polls LogiCoy



Notifications collected automatically

Notification from LogiCoy	What happens in OpenEMR
Renewal request	Added to the inbox and the patient's card, awaiting the prescriber
Change request	Added to the inbox and the patient's card, awaiting the prescriber
Cancel response	Status of the original prescription updated (cancelled, denied, not delivered)
Fill event	Fill status looked up and shown on the sent prescription
Delivery / verification	Delivery and verification state updated on the sent prescription

05 · FEATURES FOR ADMINISTRATORS

Practice administration and onboarding

Everything needed to set up a practice happens inside OpenEMR. There is no separate portal to learn and no data to key in twice.

Register from OpenEMR

Register the practice, each service location and each prescriber with LogiCoy from the module's configuration screen, using details already in OpenEMR's facility and user records.

Prescriber eRx profile on the user screen

An eRx block on **Administration → Users** captures what registration needs (date of birth, gender, prescribing type, address) next to the core NPI and DEA fields.

eRx or EPCS per prescriber

Choose **eRx** or **EPCS** for each prescriber. EPCS registration checks that the DEA number, DEA expiry, DEA state, date of birth and SSN last 4 are all present. If any are missing, it names them and registers nothing.

Encrypted credential store

Load your LogiCoy signing key by upload, by paste or by server path. It is encrypted with OpenEMR's own key material and never shown again.

Multi-facility, multi-prescriber

One credential per OpenEMR facility, with an optional per-prescriber credential. The prescriber's own credential is used when present, otherwise the facility's.

Credential lifecycle

Rotate a client ID without touching the key, replace the key by uploading a new one, or deactivate a credential. The list shows whether a key is present, never the key itself.

Settings in OpenEMR Globals

Administration → Globals → LogiCoy eRx holds the on/off switch, the backend URL and inbound polling. Secrets are never stored in Globals.

Prescription log for support

Modules → LogiCoy eRx Log shows queue activity: direction, type, message ID, status, attempts and last error. Prescription content is not shown, so it is safe to share with support.

Where each setting lives

Screen	Purpose	Permission required
Patient Summary card, prescribing screen, inbox	Prescribe, answer pharmacy requests, check status	Same as OpenEMR's native Rx (`patients / rx`)
Modules → Manage Modules → LogiCoy eRx	Connection settings, credentials, practice / location / prescriber registration	Administrator (`admin / super`)
Administration → Users	Prescriber eRx profile	User administration (`admin / users`)
Administration → Globals	Module switches and backend URL	OpenEMR Globals access

06 · UNDER THE HOOD

Data flow, sync and reliability

Most practices run OpenEMR behind a firewall with no inbound access. The module is built for that: **it only ever calls out**. LogiCoy never needs to connect in.

Sending: a prescription leaves the chart



Receiving: pharmacy traffic comes back



Built to be reliable

Screens never wait on the network

Cancels and pharmacy responses are queued and sent in the background, so a slow connection never freezes a clinician's screen. Queued items show as *queued* until sent.

No duplicates

Every message carries a unique LogiCoy message ID, and the database enforces it. If cron fires twice or a poll repeats, nothing is sent or written twice.

Safe to run concurrently

Workers claim queue rows before sending them, so overlapping cron runs do not double-send. Failed sends are retried, and the attempt count and last error are kept.

OpenEMR data the module uses

OpenEMR data	Access	Why
Patient demographics	Read	Identify the patient to LogiCoy and the pharmacy
Users (prescribers)	Read / write	NPI, DEA and contact details for registration. Writes only the address and phone fields an admin edits on the user screen.
Facilities	Read	Practice and service-location registration
Audit log	Write	One audit event per prescription sent, under the signed-in user
Module tables `logicoy_erx_*`	Read / write	Credentials, queue, message map, patient map, prescriber profile, prescription tracking, notifications

07 · TRUST

Security, privacy and compliance

Prescribing is an audited clinical action that uses sensitive credentials. The module protects both, and uses OpenEMR's own security features rather than adding its own.

Credentials and keys

- Signing keys are encrypted at rest with OpenEMR's built-in encryption (`CryptoGen`)
- Uploaded keys are stored at mode `0600` in OpenEMR's certificates directory, which the web server will not serve over HTTP
- A server-path key is refused if it is missing, readable by other users, not an RSA private key, or anywhere the web server could serve it
- Keys are never written to the module directory, never logged, never shown again
- Secrets are kept out of OpenEMR Globals, which every administrator can read and which appear in backups

Authentication to LogiCoy

- OAuth 2.0 access tokens obtained with a signed JWT client assertion (RS256) made from your practice key
- Your private key never leaves your server; only signed assertions are sent
- All traffic is HTTPS and outbound only; no inbound ports are opened
- Per-practice and optional per-prescriber credentials

Access control

- Prescribing screens use the **same permission as OpenEMR's native Rx**
- Credential and registration screens require the administrator permission
- Pharmacy requests can only be answered by the prescriber they are addressed to
- Every form and API action is protected by OpenEMR's CSRF tokens

Audit and PHI handling

- Every prescription sent is recorded in OpenEMR's audit log (`EventAuditLogger`) under the signed-in user
- The support log shows message IDs and status only, never prescription content
- The SSN fragment used for EPCS is encrypted at rest and never shown after saving
- Rejected form fields are logged by field name only, never by value

Compliance posture

Area	How it is met
HIPAA	LogiCoy's platform is fully HIPAA compliant. The module keeps PHI inside OpenEMR and sends only what a prescription needs, over encrypted connections.
EPCS	Signing and identity proofing run on LogiCoy's EPCS-certified platform. The module only collects the prescriber's registration details and checks they are complete.
Regulatory changes	Message formats and routing rules are maintained centrally by LogiCoy, so practices get changes without upgrading the module.
Open source licensing	The module is distributed under GPL v3, the same licence as OpenEMR. It adds no third-party runtime dependencies and uses only libraries OpenEMR already ships.

08 · FOR YOUR IT TEAM

Technical requirements and compatibility

System requirements

Requirement	Detail
OpenEMR	7.0.0 or later. Verified against OpenEMR 8.2.0.
PHP	8.1 or later
Database	OpenEMR's existing MySQL / MariaDB database. The module adds its own tables, all prefixed `logicoy_erx_`.
Network	Outbound HTTPS (TCP 443) from the OpenEMR server to the LogiCoy backend. No inbound connection, VPN or port forwarding needed.
Scheduler	A working cron (or equivalent) on the OpenEMR server to run the queue worker, typically every 5 minutes per facility
Server access	Root or sudo to run the installer, plus an OpenEMR administrator account (`admin / super`)
Hosting	Bare metal, virtual machine or Docker. The installer detects an OpenEMR container and installs into it.

Compatibility matrix

Module version	OpenEMR versions	Notes
1.0.0 – 1.0.5	7.0.0 – 7.0.3	Developed and tested against 7.0.3
1.0.6 – 1.0.7	7.0.0 – 8.2.0	Verified against 8.2.0, including OpenEMR 8's new session handling. 1.0.7 adds cancel, renewal, change and status workflows.

Integration points in OpenEMR

OpenEMR surface	Used for
Patient Summary render event	The LogiCoy eRx card in patient context
Menu and Globals events	Inbox and Log menu entries, and the module settings section
User create / edit events	The prescriber eRx profile block on Administration → Users
Audit logger	Recording each prescription sent in OpenEMR's audit log
Module lifecycle listener	Install, enable and upgrade actions from Manage Modules

09 · OPERATING MODEL

Deployment, upgrades and responsibilities

The module runs on your OpenEMR server; the eRx platform runs at LogiCoy. This page shows how the module is installed and maintained, and who looks after each part.

Installation footprint

Single self-extracting installer

LogiCoy supplies one file, `install-oe-module-logicoy-erx-<version>.sh`. It finds OpenEMR, checks the OpenEMR and PHP versions, places the module and sets file ownership. It supports `--dry-run`, a custom OpenEMR path, Docker containers and an extract-only mode for configuration-management deployments.

Standard OpenEMR module

Installs to `interface/modules/custom_modules/` and is registered, installed and enabled from **Modules → Manage Modules**. It changes no OpenEMR core files and adds no vendor packages.

Safe upgrades

Upgrading archives the previous version first and leaves the module's tables and credentials alone. Numbered database migrations run automatically, and install can be re-run safely.

Clean uninstall

Disabling or unregistering keeps the module's tables. Prescription records are audited clinical data, so removing them is a deliberate manual step, done with LogiCoy support.

Who does what

Responsibility	Practice	LogiCoy
OpenEMR server: hosting, backups, OS and database maintenance	Owns	–
OpenEMR upgrades, checked against the compatibility matrix	Owns	Advises
Module install and upgrades	Runs the installer	Supplies installer and guidance
Queue worker: cron schedule and monitoring	Owns	Advises
Signing key custody	Stores securely	Issues and rotates on request
Prescriber and facility data: NPI, DEA, addresses	Keeps accurate	Validates at registration
eRx platform: drug data, routing, EPCS certification, regulatory updates	–	Owns
Support: setup questions, message tracing, pharmacy issues	Shares the message ID from the log	Toll-free support

10 · GETTING STARTED

Implementation and go-live

Nine steps from receiving your onboarding pack to the first prescription. Each one is covered in detail in the Client Setup Guide.

1	**Install the module files** Run the LogiCoy installer on the OpenEMR server.	Server shell System administrator
2	**Register, install and enable** Modules → Manage Modules → LogiCoy eRx → Register, Install, Enable.	OpenEMR Administrator
3	**Switch the module on** Administration → Globals → LogiCoy eRx → Enable.	OpenEMR Administrator
4	**Connection settings** Enter the backend URL supplied by LogiCoy.	Module config Administrator
5	**Load the practice credential** Client ID, client registration UUID and private signing key, per facility.	Module config Administrator
6	**Set up each prescriber** Complete the LogiCoy eRx block on each prescriber's user record; choose eRx or EPCS.	Administration → Users Administrator
7	**Register with LogiCoy** Register the practice, service location and prescribers from the config screen.	Module config Administrator
8	**Schedule the queue worker** One cron line per facility, every 5 minutes.	Server cron System administrator
9	**Verify** Open a patient chart, confirm the LogiCoy eRx card, check the log.	Patient chart Any prescriber

LOGICOY SUPPLIES

- The installer file for your module version
- Backend base URL
- Client ID and eRx client registration UUID
- Private signing key (`.pem` / `.key`): keep it secure and never email it
- Onboarding help and toll-free support

YOUR PRACTICE PROVIDES

- An OpenEMR server that meets the requirements on page 10
- Server (root / sudo) and OpenEMR administrator access
- Accurate facility records: address, phone, NPI
- For each prescriber: NPI, email, date of birth, address; for EPCS also DEA details and SSN last 4

11 · REFERENCE

Feature summary

Capability	Detail	Where
Launch in patient context	Patient Summary card, permission-gated	Module
Drug search	By product name or NDC, controlled substances flagged	Module + platform
Pharmacy search	By name, city or ZIP	Module + platform
New prescriptions	Drafts, edit, remove, send NewRx	Module + platform
EPCS	Per-prescriber type, completeness checks, certified signing	Module + platform
Renewal requests	Approve, or deny with reason code	Module + platform
Change requests	Approve, validate or deny	Module + platform
Cancel	From the sent list, response tracked	Module + platform
Status tracking	Prescription, fill and verification status	Module + platform
Pharmacy inbox	Practice-wide list of pending requests	Module
Audit trail	Every send in OpenEMR's audit log; support log without PHI	Module
Onboarding	Practice, location and prescriber registration from OpenEMR	Module + platform
Medication history, eligibility, ePA, RTPB	Available on the LogiCoy platform; ask LogiCoy about availability in OpenEMR	Platform

Scope notes for this release

- The module works **alongside** OpenEMR's native prescription screen. Prescriptions entered in the native screen are not sent through LogiCoy.
- Renewal and change responses of type *approved with changes* or *replace* are not yet offered in the module. If a request allows only those, the screen says so.
- Prescriptions sent through LogiCoy are tracked on the patient's LogiCoy eRx screen and in the audit log. They are not yet written into OpenEMR's native prescription list.
- Pharmacy requests appear in the LogiCoy eRx Inbox and on the patient card, not in OpenEMR's core Messages.
- Pharmacy traffic arrives on the worker's schedule, so a new request or reply can take up to one worker interval (typically 5 minutes) to appear.

LogiCoy

A team that actually picks up the phone.

Questions about deployment, credentials or a prescription in the log? Contact LogiCoy support and share the message ID from the LogiCoy eRx Log.

TOLL-FREE

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BOOK A DEMO

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